

Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The purpose of this document is to define the requirements for the Airport Management System (AMS). The system is designed to support airport operations and improve the interaction between passengers, airport staff, airline companies, and administrators.

This document specifies the system modules, actors, use cases, functional requirements, and non-functional requirements that describe how the system should operate.

The document serves as the foundation for later stages of the software engineering process, including system design, modeling, and development.

1.2 System Overview

Airports are complex environments where many operations must be coordinated simultaneously. These operations include passenger services, flight scheduling, baggage handling, security control, staff management, financial transactions, and airport resource management.

The Airport Management System (AMS) is an integrated software platform that manages these operations through a centralized digital system.

The system helps:

- Passengers access travel services
- Airport staff manage operations
- Airlines manage flights
- Administrators monitor system activities

The system improves efficiency, accuracy, and the overall airport experience.

1.3 System Scope

The Airport Management System (AMS) is designed to support and manage major airport operations through a centralized software platform. The system integrates multiple operational modules including passenger services, flight operations, baggage management, airport resources, security, finance, human resources, transportation services, and system administration. The system allows passengers to search and book flights, perform online check-in, track baggage, and receive flight notifications. Airport and airline staff can manage operational activities such as scheduling flights, assigning gates, handling baggage, and monitoring airport facilities. The system aims to improve

operational efficiency, data accuracy, communication between departments, and the overall passenger experience at the airport.

1.4 Assumptions

- ✓ Users have access to an internet connection in order to interact with the system.
- ✓ Passengers must create user accounts before booking tickets or performing online check-in.
- ✓ Airlines provide accurate flight schedules and operational data.
- ✓ Airport staff are properly trained to operate the system modules.
- ✓ The system will be accessed primarily through web-based interfaces.
- ✓ The system database will store passenger, flight, baggage, and operational information.

1.5 Constraints

- ✓ The system must comply with aviation security and data protection regulations.
- ✓ The system must operate using a centralized relational database.
- ✓ The system must be accessible through modern web browsers.
- ✓ The system must ensure data privacy and protection of passenger information.
- ✓ The system must integrate with existing airport information systems where required.

1.6. Future Enhancements

Future versions of the Airport Management System may include additional advanced features such as:

- ✓ Mobile application support for passengers and airport staff.
- ✓ Biometric passenger identification systems.
- ✓ AI-based flight delay prediction.
- ✓ Automated baggage tracking using IoT technology.
- ✓ Real-time airport navigation for passengers.
- ✓ Integration with airline loyalty and reward programs.

2. System Users (Actors)

The system supports multiple types of users who interact with it in different ways.

Passenger - Passengers are the primary users of the system. They interact with the system to manage travel-related services such as booking flights, checking flight status, performing check-in, and tracking baggage.

Visitor - Visitors are users who access the system without having a flight booking. They may want to check flight information or explore airport facilities.

Airline Staff - Airline staff members manage flight-related information including schedules, aircraft assignments, and flight status updates.

Airport Staff - Airport staff handle operational tasks such as baggage processing, gate assignment, and facility management.

Security Officer - Security officers ensure airport safety by verifying passenger identities, monitoring security checkpoints, and managing restricted areas.

Customer Support Agent - Customer support agents assist passengers with inquiries, complaints, and service requests.

Finance Staff - Finance staff manage financial transactions, billing processes, and financial reporting related to airport services.

Human Resource Staff - Human resource staff manage employee records, staff scheduling, and role assignments.

System Administrator - System administrators manage the overall system operation including user accounts, system permissions, and maintenance.

3. Major System Modules – User Characteristics

The Airport Management System is divided into several functional modules. Each module manages a specific group of airport operations.

3.1 Passenger Service Management Module

This module manages services directly related to passengers.

Main Responsibilities

- Flight search
- Ticket booking
- Seat selection
- Online check-in
- Boarding pass generation
- Flight notifications
- Passenger support

Actors

- Passenger
- Customer Support Agent

Example Use Cases

- Search Flight
- Book Ticket
- Select Seat
- Perform Online Check-In
- Download Boarding Pass
- Receive Notifications

3.2 Flight Operations Management Module

This module manages flight scheduling and flight operations.

Main Responsibilities

- Flight scheduling
- Flight status updates

- Aircraft assignment
- Gate assignment
- Flight delay management

Actors

- Airline Staff
- Airport Staff

Example Use Cases

- Create Flight Schedule
- Update Flight Status
- Assign Aircraft
- Assign Gate
- Manage Delayed Flights

3.3 Baggage Management Module

This module manages passenger baggage tracking and handling.

Main Responsibilities

- Baggage registration
- Baggage tracking
- Baggage loading
- Lost baggage reporting
- Baggage claim management

Actors

- Airport Staff
- Passenger

Example Use Cases

- Register Baggage
- Track Baggage
- Report Lost Baggage
- Update Baggage Location

3.4 Security Management Module

This module ensures airport safety and security.

Main Responsibilities

- Identity verification
- Security checkpoint monitoring
- Security incident reporting
- Restricted area control

Actors

- Security Officer

Example Use Cases

- Verify Passenger Identity
- Monitor Security Checkpoints
- Record Security Incidents

3.5 Airport Resource Management Module

This module manages airport infrastructure and resources.

Main Responsibilities

- Gate allocation
- Runway scheduling
- Terminal management
- Equipment monitoring

Actors

- Airport Staff

Example Use Cases

- Assign Gates
- Schedule Runway Usage
- Monitor Terminal Capacity

3.6 Finance and Billing Module

This module manages financial transactions related to airport services.

Main Responsibilities

- Ticket payments
- Service billing
- Financial reporting
- Transaction tracking

Actors

- Passenger
- Finance Staff

Example Use Cases

- Process Ticket Payment
- Record Payment Transactions
- Generate Financial Reports

3.7 Human Resource Management Module

This module manages airport employees and staff operations.

Main Responsibilities

- Employee records
- Staff scheduling
- Role assignment
- Staff management

Actors

- HR Staff
- System Administrator

Example Use Cases

- Register Employee
- Assign Staff Roles
- Manage Staff Schedules

3.8 Customer Support Module

This module handles passenger support services.

Main Responsibilities

- Inquiry management
- Complaint management
- Assistance services
- Feedback handling

Actors

- Passenger
- Customer Support Agent

Example Use Cases

- Submit Complaint
- Request Assistance
- Track Support Requests

3.9 Transportation and Parking Module

This module manages airport transportation services.

Main Responsibilities

- Parking management
- Taxi services
- Shuttle services
- Transportation information

Actors

- Passenger
- Visitor

Example Use Cases

- View Parking Availability
- Book Parking Space
- View Transportation Options

3.10 System Administration Module

This module manages the system infrastructure.

Main Responsibilities

- User account management
- System monitoring
- Permission control
- System maintenance

Actors

- System Administrator

Example Use Cases

- Manage User Accounts
- Monitor System Activity
- Update System Settings

4. Requirements

4.1 Functional Requirements organized by System Modules

4.1.1 Passenger Service Management

These requirements describe the services that the system provides directly to passengers.

1. Passenger searches flights
2. Passenger views flight schedules
3. Passenger books flight tickets
4. Passenger selects seat preferences
5. Passenger uploads travel documents
6. Passenger performs online check-in
7. Passenger downloads boarding pass
8. Passenger checks assigned gate information
9. Passenger tracks baggage status
10. Passenger receives flight notifications
11. Passenger submits customer support request
12. Passenger submits complaint
13. Passenger updates personal profile information
14. Passenger cancels flight booking
15. Passenger requests ticket refund

4.1.2 Flight Operations Management

These requirements manage flight scheduling and airline operations.

16. Airline staff creates flight schedules
17. Airline staff updates flight status
18. Airline staff assigns aircraft to flights
19. Airline staff manages flight delays
20. Airline staff cancels flights

21. Airline staff updates departure times
22. Airline staff updates arrival times
23. Airline staff assigns flight crew
24. Airline staff coordinates boarding processes
25. Airline staff monitors flight operations

4.1.3 Baggage Management System

These requirements handle passenger baggage processing and tracking.

26. Airport staff registers passenger baggage
27. Airport staff tracks baggage movement
28. Airport staff records baggage loading
29. Airport staff updates baggage location
30. Airport staff reports lost baggage
31. Airport staff manages baggage claims
32. Passenger reports delayed baggage
33. Airport staff schedules baggage delivery

4.1.4 Security Management System

These requirements support airport security and safety operations.

34. Security officer verifies passenger identity
35. Security officer monitors security checkpoints
36. Security officer records security incidents
37. Security officer manages restricted area access
38. Security officer verifies boarding pass at security checkpoints
39. Security officer manages emergency security alerts

4.1.5 Airport Resource Management

These requirements manage airport infrastructure and operational resources.

- 40. Airport staff assigns gates to flights
- 41. Airport staff schedules runway usage
- 42. Airport staff monitors terminal occupancy
- 43. Airport staff manages airport facility resources

4.1.6 Customer Support Management

These requirements handle passenger assistance and customer service operations.

- 44. Customer support agent responds to inquiries
- 45. Customer support agent tracks service requests

4.1.7 Finance and Billing Management

These requirements manage financial transactions and billing processes.

- 46. Passenger completes ticket payment
- 47. Finance staff records payment transactions
- 48. Finance staff generates billing reports
- 49. Finance staff processes service charges
- 50. Finance staff generates financial transaction reports
- 51. Finance staff verifies payment confirmations

4.1.8 Human Resource Management

These requirements support airport staff management.

- 52. HR staff registers employee records
- 53. HR staff manages employee schedules
- 54. HR staff assigns staff roles

4.1.9 System Administration

These requirements manage the system infrastructure and user control.

- 55. System administrator manages user accounts
- 56. System administrator updates airport information
- 57. System administrator monitors system usage
- 58. System administrator manages system permissions
- 59. System administrator performs system maintenance
- 60. System administrator generates system activity reports

4.2 Non-Functional Requirements (50)

Non-functional requirements describe how the system (Airport Management System) should perform.

4.2.1 Product Requirements

Product requirements describe how the system behaves and its quality attributes (usability, performance, security, availability).

4.2.1.1 Usability Requirements

1. The system interface must be intuitive.
2. The system must allow easy navigation.
3. Users must complete common tasks with minimal steps.
4. The system must provide clear instructions.
5. The system must display helpful error messages.
6. The system must provide user help documentation.
7. The interface must be visually consistent.
8. The system must be accessible to first-time users.

4.2.1.2 Performance Requirements

9. The system must load pages within three seconds.
10. The system must process searches within two seconds.
11. The system must support at least 1000 concurrent users.
12. Payment processing must complete within five seconds.
13. The system must support at least 1500 simultaneous users during peak travel periods.

14. The system must optimize database queries.
15. The system must maintain stable performance under high workload conditions.
16. The system must process requests without noticeable delays.

4.2.1.3 Availability

17. The system must operate 24/7.
18. System downtime must not exceed 1% of total operational time.
19. Maintenance must occur during low-traffic hours.

4.2.1.4 Security

20. The system must require user authentication.
21. Passwords must be encrypted.
22. The system must protect personal data.
23. The system must prevent unauthorized access.
24. Sessions must expire after inactivity.
25. Login attempts must be logged.
26. Data transmission must use secure communication protocols.
27. Sensitive information must be protected.

4.2.2 Organizational Requirements

These requirements relate to how the system is developed, maintained, and structured.

Maintainability

28. The system must have modular code.
29. The system must be well documented.
30. Updates must be easy to apply.
31. Bugs must be easy to identify and fix.
32. Developers must be able to add new features easily.

Reliability

33. The system must maintain at least 99% system uptime.
34. The system must recover from minor failures.
35. The system must maintain data integrity during all transactions.

- 36. The system must prevent data corruption.
- 37. Regular system backups must occur.
- 38. System errors must be logged.

Data Management

- 39. User data must be stored securely.
- 40. Flight data must remain accurate.
- 41. The system must maintain data consistency across all system modules.

4.2.3 External Requirements

These requirements concern interaction with external environments (devices, systems, platforms).

Compatibility

- 42. The system must support major web browsers.
- 43. The system must support mobile devices.
- 44. The system must support tablets.
- 45. The system must operate on multiple operating systems.
- 46. The interface must adapt to different screen sizes.

Scalability / Integration

- 47. The system must support future increases in the number of users.
- 48. The system must support future system expansion.
- 49. The system must allow integration with other systems.
- 50. The system must handle large volumes of operational and passenger data efficiently.

5. User Scenarios / Use Cases

Use Case 1: Passenger Books a Flight Ticket

Actor: Passenger

Description:

A passenger searches for available flights, selects a preferred flight, enters passenger information, and confirms the booking.

Steps:

1. Passenger logs into the system.
2. Passenger searches for available flights.
3. Passenger selects a flight.
4. Passenger enters booking information.
5. Passenger confirms the ticket booking.

Outcome:

The system stores the booking information and confirms the ticket.

6. Conclusion

The Airport Management System integrates multiple airport operations into a unified digital platform. The requirements described in this document define the system's expected behavior, operational capabilities, and quality attributes.

These requirements will guide future development phases including system modeling, architectural design, database design, and implementation.